Maria Camarca

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Education

2019 – 2025 **Ph.D. Planetary Science**
California Institute of Technology, Pasadena CA
Thesis: Fresh Eyes for an Old Moon: ALMA and JWST Perspectives of Callisto
Advisor: Prof. Katherine de Kleer

2019 – 2021 **M.S. Planetary Science**
California Institute of Technology, Pasadena CA

2017 – 2019 **Part-time coursework in math & physics**
American University, Washington D.C.

2013 – 2017 **B.S. Biology & minor in Psychology**
Marymount University, Arlington VA

Research Positions

2026 – **NSF Astronomy & Astrophysics Postdoctoral Fellow**
National Radio Astronomy Observatory, Charlottesville VA

2025 – 2026 **Postdoctoral Scholar Research Associate**
2019 – 2025 **Graduate Student**
California Institute of Technology, Pasadena CA

2017 – 2019 **Research Assistant**
2017 **Undergraduate Research Associate in Astrobiology**
NASA Goddard Space Flight Center, Greenbelt MD
Contract through American University, Washington D.C.

Honors

2026 NSF Astronomy & Astrophysics Postdoctoral Fellowship

2025 Caltech Graduate Studies Office Excellence in Leadership and Service Award

2022 – 2026 Keck Institute for Space Studies Affiliate

2021 NSF Graduate Research Fellowship

2013 – 2017 Marymount University Presidential Scholarship

Publications

In prep **Camarca, M.**, Lunine, J., Trumbo, S., et al. "A global surface component on Titan as observed with JWST".

First-author

5. **Camarca, M.**, de Kleer, K., Cartwright, R. J., et al. 2026. "Callisto from JWST: CO_2-rich Terrain on the Leading Hemisphere and Global Patterns of H_2O Ice". *The Planetary Science Journal*, Vol. 7, No. 9.
4. **Camarca, M.**, de Kleer, K., Butler, B., et al. 2025. "A Multifrequency Global View of Callisto's Thermal Properties from ALMA". *The Planetary Science Journal*, Vol. 6, No. 8.
3. **Camarca, M.**, de Kleer, K., Butler, B., et al. 2023. "Thermal Properties of the Leading Hemisphere of Callisto Inferred from ALMA Observations". *The Planetary Science Journal*, Vol. 4, No. 8.

2. **Camarca, M.**, Murphy, B., and Jaber, D. 2022. "The Evolution of Games and Case Studies in Organic Chemistry". *The Chemical Educator*, Vol. 27, No. 1.

1. **Camarca, M.**, Heuett, W., and Jaber, D. 2019. "CHEMCompete-II: An Organic Chemistry Card Game To Differentiate between Substitution and Elimination Reactions of Alcohols". *Journal of Chemical Education*, Vol. 96, No. 11.

Co-author

11. Bézard, B., [...], **Camarca, M.**, et al. 2026. "An unidentified absorption feature at 5.11 μm on the surfaces of Titan and Pluto from JWST spectroscopy". *Astronomy & Astrophysics,* Vol. 711.

10. Meyer, C., **Camarca, M.**, et al. 2026. "Revealing Callisto's Near-subsurface Thermophysical Properties with ALMA Calibration Data". *The Planetary Science Journal*, Vol. 7, No. 1.

9. Milby, Z., de Kleer, K., Schmidt, C., and **Camarca, M.** 2026. "Variability of Europa's Optical Aurora on Orbital and Multiyear Timescales". *The Planetary Science Journal*, Vol. 7, No. 3.

8. Cartwright, R. J., [...], **Camarca, M.**, et al. 2024. "Revealing Callisto's Carbon-rich Surface and CO_2 Atmosphere with JWST". *The Planetary Science Journal*, Vol. 5, No. 3.

7. Thelen, A. E., [...], **Camarca, M.**, et al. 2024. "Subsurface Thermophysical Properties of Europa's Leading and Trailing Hemispheres as Revealed by ALMA". *The Planetary Science Journal*, Vol. 5, No. 2.

6. de Kleer, K., [...], **Camarca, M.**, et al. 2023. "The Optical Aurorae of Europa, Ganymede, and Callisto". *The Planetary Science Journal*, Vol. 4, No. 2.

5. Pereira, C. L., [...], **Camarca, M.**, et al. 2023. "The two rings of (50000) Quaoar". *Astronomy and Astrophysics*, Vol. 673.

4. Faggi, S., [...], **Camarca, M.**, et al. 2021. "The Extraordinary Passage of Comet C/2020 F3 NEOWISE: Evidence for Heterogeneous Chemical Inventory in Its Nucleus". *The Astronomical Journal*, Vol. 162, No. 5.

3. Shepard, M. K., [...], and **Camarca, M.** 2021. "Asteroid 16 Psyche: Shape, Features, and Global Map". *The Planetary Science Journal*, Vol. 2, No. 4.

2. Lippi, M., [...], **Camarca, M. N.**, et al. 2020. "New Insights into the Chemical Composition of Five Oort Cloud Comets after Re-analysis of Their Infrared Spectra". *The Astronomical Journal*, Vol. 159, No. 4.

1. Paganini, L., **Camarca, M. N.**, et al. 2019. "Observations of Jupiter Family Comet 252P/LINEAR During a Close Approach to Earth Reveal Large Abundances of Methanol and Ethane". *The Astronomical Journal*, Vol. 158, No. 3.

Recent Presentations

Feb 2026	Pomona College Physics Colloquium | Claremont CA
Sep 2025	DPS-EPSC Joint Meeting (talk) | Helsinki, Finland
Aug 2025	JHU-APL Brown Bag Seminar Seminar (invited) | Laurel MD
Oct 2024	56th AAS Division of Planetary Science meeting (talk) | Boise ID
Oct 2023	55th AAS Division of Planetary Science meeting (talk) | San Antonio TX

Jan 2023	NRAO/UVA Tuna Talk (invited) | Charlottesville VA
Oct 2022	54th AAS Division of Planetary Science meeting (talk) | London, Ontario
Oct 2021	53rd AAS Division of Planetary Science meeting (virtual poster)

Telescope Programs

ALMA	**2** co-I programs: Cycle 11 PI: A. Akins, Cycle 11 PI: A. Thelen
JWST	**1** co-I program: Cycle 3 (MIRI) PI: R. Davis
Palomar	**1** co-I program: 2022A (CHIMERA) PI: K. de Kleer
Keck	**1** Science-PI program: 2020B (NIRSPEC) PI: G. Blake **2** co-I programs: 2021B, 2022A (HIRES) PI: K. de Kleer

Teaching Experience

2021 – 2024	**Teaching Assistant** | **Caltech Planetary Science Department** Ge 103/11c: Introduction to the Solar System
2021 – 2022	**RISE Program Tutor** | **Caltech Y** Elementary and high school math and biology
2014 – 2017	**Supplemental Instructor/General Tutor** | **Marymount University** General and organic chemistry

Mentoring Experience

2024, 2023	Cole Meyer (Princeton undergraduate) | Caltech SURF program
2021	Lily Coffin (Caltech undergraduate) | Caltech SURF program
2021	Tyler Mapes (Caltech undergraduate) | Caltech SURF program

Service and Outreach

Jun 2026	Pasadena City of Astronomy Festival | *volunteer*
Mar 2026	Caltech Astronomy Stargazing Lecture | *panelist*
Apr 2026	Mary W. Jackson Elementary STEAM Night, Altadena CA | *science demo*
2021-2026	Union Station Homeless Services, Pasadena, CA | *11× dinner lead*
Aug 2025	Old Town Monrovia Sidewalk Astronomy | *telescope guide*
Sep 2024	Caltech Keck Remote Observing Room | *tour for physics/astronomy teachers*
Apr 2024	Caltech Astronomy Stargazing Lecture | *panelist*
Feb 2024	Astronomy on Tap, Pasadena CA | *pub outreach talk*
Feb 2024	STEAM Night Highland Oaks Elementary, Arcadia CA | *outreach talk*
Jul 2023	Flintridge Center: Youth of Promise program, Pasadena CA | *outreach talk*
Jul 2023	Family Promise Homeless Ministry, St. Philip, Pasadena CA | *overnight host*
2021-2022	Caltech Dix Planetary Science Seminar | *co-organizer*